

The Relational Coarse-Graining Framework, Version 8.0

The Frontier Programme: certified emergence, fusion, and phase-coded back-coupling on a two-channel spin-1 substrate. Twelve machine-certified rounds (R39–R50), thirteen verification interludes (V-1–V-13), ten sealed predictions, thirteen laws with falsifiers.

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Supersedes RCF-FW-v7.0 (post-arc specification). Verified-change basis: the frontier-programme artifact chain, enumerated in Appendix C of this document.

Keywords — relational coarse-graining · open quantum systems · emergent observables · Lindblad dynamics · interference carriers · back-coupling · pre-registered verification

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Abstract

This document is Version 8.0 of the Relational Coarse-Graining Framework (RCF), written for the first time as a formal physics paper and superseding the specification RCF-FW-v7.0. It consolidates, in one self-contained and mathematically complete account, every verified change registered by the *frontier programme* — the twelve-round machine-certified investigation (rounds R39–R50, verification interludes V-1–V-13) of a two-channel open spin-1 substrate. The programme asked a single question in ascending levels of specificity: does the interface between the substrate’s algebraic tier and its dynamical tier — the *frontier* — exist, and if so, what is it?

The certified answer, built round by round, is a complete causal loop with every segment machine-measured. The static tier possesses a composite frontier with a mesoscopic corner: the deficit law $D(\alpha, N) = C(N) \alpha^{\beta(N)}$ with $\beta \rightarrow 1.95$, the curvature law $\kappa(N) \propto N^{-\gamma_C}$ with $\gamma_C = 0.6131 \pm 0.0187$, corner width $\sim N^{0.31}$, and a band-closing law $e^{-N/1.05}$ that the curvature provably does not echo (κ/ε grows by a factor 240 across the ladder). The margin — the escape of the joint dynamics above the static frontier — decomposes into four certified carriers, $\Delta = C + L + J + \chi$ with $\chi = +0.0062$ a constructive interference term absent from every upstream ledger on three independent routes: the crossing *de-factorises* the two channels and produces χ as the connected residue of the joint measure it constructs. χ is thereby certified *constitutively native* — the framework’s first observable whose existence is an act rather than a property. Its decay follows the summed-exponent law $p_\chi = p_L + p_J = -1.861$, which then holds on two further external objects (the corner’s stress response and the exclusion chart’s shift) and carries an explicit falsifier. The produced object couples back into the substrate: the kernel’s response is phase-coded, $\delta K(\sigma, \tau) = \Lambda R(\sigma, \tau - \tau_{\text{mem}})$, with lag $\tau_{\text{mem}} = \tau_0 N^{0.31} \gamma^{-0.743}$, cancellation gain $\Lambda \sim \gamma^{+1.861}$, and a γ -flat product — exponent conservation at the interface, the framework’s first original conservation law. The loop closes in the substrate’s own documents: the exclusion boundary $\mu_1^2 = a\Gamma + b$ carries a localized inscription band at the corner’s footprint, a 1.7% order-sensitive shift addressed at 50:1 density and transcribed by a flat dictionary. The response space of the order memory is a two-phase torus, $R(\sigma) = 2|\chi| u(\theta_L) \otimes u(\theta_J)$, with unit-modulus phases: one bit of amplitude that is an arithmetic identity, and a full vector of direction. Every result is held at a declared certification grade under a pre-commitment apparatus that sealed ten predictions (eight confirmed, one killed, two pending), closed the register’s second question by sealed rule rather than threshold, and maintained a three-family defect taxonomy stable for ten consecutive rounds. The framework’s compressed conclusion: the frontier is not a separator but the organ the substrate uses to talk to itself, in the phase language of its own channels, with interference arithmetic, on the clock of its slowest mode, written where it lives.

Document lineage: RCF-FW-v7.0 $\rightarrow$ **RCF-FW-v8.0** (this document). Verified-change basis: rounds R39–R50 and interludes V-1–V-13 of the frontier programme; the full artifact chain (scripts, checkpoints, run logs, certified backups) is enumerated in Appendix C.

1 Introduction

1.1 From black-hole information to relational coarse-graining

The Relational Coarse-Graining Framework was inaugurated to address a question whose precise formulation has shifted across three decades of black-hole physics: how does information survive the strain that a strong field imposes on the Hilbert space that describes it? The framework’s founders declined the two standard exits — the firewall answer, which makes survival a non-event by discarding the interior, and the unitarity-by-decoherence answer, which salvages the global unitary but loses the operational meaning of local observables under strain. Both exits avoid the question; RCF attempts to answer it. The framework’s central wager is that the right level of description is neither the local operator algebra of an effective field theory nor the global state of a closed system, but a *relational coarse-graining layer* in between: a decomposition of the operator algebra into equivalence classes respecting a designated notion of locality and symmetry, together with the coarse-grained observables those classes generate. The wager is that this layer, properly constructed, supports a notion of robustness under strain that neither the local nor the global description can provide [?].

Across its development the framework matured through four phases: the operator-algebraic foundations with the verified-number registry and the protocol system (v1–v4); the epistemic hardening with the Type-A/Type-B classification, the Hessian identity, and the status vocabulary (v6.0–v6.5); the six-round AKLT substrate arc, which closed with a negative verdict on every dynamical form of the projection picture that the substrate could host (v7.0); and the phase this document closes: the *frontier programme*, a twelve-round machine-certified investigation of the interface between the algebraic and dynamical tiers of a two-channel open spin-1 substrate. Version 7.0 ended by launching a substrate-change fork whose selection criterion was the primitive-vs-emergent axiom’s operational consequence. The frontier programme is that fork’s execution: rather than changing the substrate outright, it changed the *question* — from “can the projection picture survive here?” to “what does the interface between the substrate’s two tiers actually do?” — and equipped the change with an instrument discipline (pre-registered sheets, sealed predictions, byte-level verification interludes) that makes every answer in this paper reproducible from the artifact chain on disk.

1.2 The programme and its rounds

The frontier programme comprises twelve data rounds, each followed by or paired with a machine verification interlude. The chain reads R39 → R50 with interludes V-1 through V-13 distributed throughout; every link is either machine-held (its certified checkpoint reproduces byte-exactly under fresh re-execution) or debt-cleared by its interlude before the next round fired. The rounds divide naturally into three movements. The *establishment movement* (R39–R43) built the executable substrate — the free sector battery, the two-convention vacuum fork on the interacting spin-1 chain, the composite-gap α -scan, the interacting frontier cells — and culminated in the frontier property census, which certified the margin field, the corner geometry, the approach law, and the dial response family. The *composition movement* (R44–R47) ran the N-ladder (curvature law, the two zeros, the order memory), the decisive N=14 rung, the carrier decomposition, and the nativity test. The *closure movement* (R48–R50) measured the return path, the inscription, and the consolidation of the register’s own most load-bearing object. Table ?? in Appendix C summarizes the chain with each round’s certified verdicts.

The paper follows the programme’s own order, because the order is load-bearing: each round’s instruments are certified before the next round imports them, and every number quoted below regenerates under a named extractor at a declared tolerance class. This is not a stylistic preference but the framework’s epistemic core — Protocol #15 (independent recomputation) applied at the level of the whole programme.

1.3 Contributions

The verified changes consolidated by this document, in the order the register holds them:

1. **The static frontier and its mesoscopic corner (§5).** The composite anatomy $\mathfrak{M}(\alpha) = \mathfrak{M} + \alpha\hat{N}$ obeys the deficit law $D = C(N)\alpha^{\beta(N)}$ with β drifting from 1.9758 to 1.9509 across $N = 6\text{--}14$; the curvature $\kappa(N) = 1/(2C)$ decays as a power law $\kappa \propto N^{-\gamma_C}$ with $\gamma_C = 0.6131 \pm 0.0187$ — neither the edge class (0) nor the bulk class (1) but a certified *mesoscopic* object of width $\sim N^{0.31}$, provably not the band edge’s echo (κ/ε grows $\times 240$).
2. **The three-way zero ordering (§5.4).** On the γ dial the projected zero, the margin zero, and the transport zero separate by three orders of magnitude: $\gamma_{\text{proj}} \in (30, 100) < \gamma_{\Delta} = 2.6 \times 10^5 < \gamma_T = 5.9 \times 10^9$; the margin dies strictly faster than its transport driver (ratio exponent -0.9606 ± 0.0166), and the decomposition-completeness fork is thereby adjudicated: *non-transport carriers exist*.
3. **Fusion and the carrier χ (§7).** The margin decomposes as $\Delta = C + L + J + \chi$ with $\chi = +0.0062$ (20.5%) a constructive cross-term confirmed by the sealed additivity null — the census’s first certified *combination* event; the corner’s mass fraction $C/\Delta = 0.551$ is simultaneously the measured growth lever of the N-side turn, an identity closed arithmetically.
4. **Constitutive nativity (§8).** χ has no upstream face on three independent routes (algebraic, correlator, simulation): the crossing *de-factorises* the channels, constructing the joint measure whose connected residue χ is. The upstream lacks not the value but the capacity.
5. **The summed-exponent law and its travels (§8.4).** Interference carriers inherit the summed decay of their parents, $p_{\chi} = p_L + p_J = -1.861$; the law is certified on two further external objects (stress response, chart shift) and carries a falsifier aimed at every future carrier of its class.
6. **The return path (§9).** The kernel listens: $\delta K = \Lambda R(\tau - \tau_{\text{mem}})$ with lag $1.02 \tau_{\text{mem}}$, gain $\Lambda \sim \gamma^{+1.861}$, γ -flat product over a $\times 7.7$ decay of the source, and byte-level upstream absence — *exponent conservation at the interface*, the framework’s first original conservation law.
7. **The response torus (§9.2).** The order memory’s response space is $R(\sigma) = 2|\chi| u(\theta_L) \otimes u(\theta_J)$ with $|u| = 1.0000 \pm 0.0003$ on all 639 enumerated cells: two phase dials, no amplitude dial — the memory remembers *which way*, never *how much*.
8. **The inscription (§10).** The substrate’s exclusion boundary is a palimpsest: the autonomous line $\mu_1^2 = a\Gamma + b$ plus a localized band at the corner’s footprint carrying a 1.7% order-sensitive shift, addressed at 50:1 density, decaying at $\gamma^{-1.861}$, transcribed by the flat dictionary $\delta\mu_1 = c(2\chi)$, $c = 0.94 \pm 0.03$. The loop closes in the documents, where phase becomes geometry.
9. **Consolidation (§11).** Under certified corner stress the four corner laws fail together (rank-1 response matrix; 55% identity stress-covariant): the corner is one object. The band is the corner’s address carrying the channel’s content.
10. **The apparatus itself (§4).** Ten sealed predictions (eight confirmed, one killed at sealed scope with its salvage quarantined under a fresh falsifier, two pending); the register’s second question closed by sealed rule rather than threshold; a three-family defect taxonomy stable for ten consecutive rounds; and the crossover-difference census (Classes F0–F6) classifying every registered observable by what crossing the frontier does to it.

1.4 Organization

Section 2 compresses the inherited framework: axioms, the Hessian identity, the robustness principle, the epistemic classification, and the AKLT arc’s negative verdict. Section 3 defines the

frontier substrate — the two books, the two channels, the crossing, the dials, the two frontiers — with all operator-level detail needed to regenerate every number in this paper. Section 4 specifies the verification apparatus. Sections 5–11 present the programme’s results in execution order. Section 12 assembles the loop, the law register, and the census; Section 13 states the open problems; Section 14 concludes. Appendices A–D carry the notation, the prediction ledger, the verification interludes, and the certified value tables.

2 Foundations of the RCF (inherited)

This section compresses, without mathematical change, the framework content established through v7.0 [?]: the axioms, the Hessian identity, the hierarchical robustness principle, the epistemic classification, and the six-round AKLT arc whose negative verdict launched the substrate-change fork that the frontier programme executed. Full statements and proofs are preserved verbatim in the audit-trail artifacts; nothing in this section is altered by the frontier programme, and everything after it is.

2.1 Primitive postulates and the structural axiom

S1 (Hilbert space).

The framework operates on a finite-dimensional complex Hilbert space $\mathcal{H}$; all operator-algebraic constructions live in $B(\mathcal{H})$. Finite dimensionality is a regularization: theorems are proven first at finite N , thermodynamic limits taken where they exist. $\mathcal{H}$ carries no primitive observable structure — substrate operators are supplied, not framework primitives.

S2 (Spectrum).

The framework takes as input a Hamiltonian H with spectrum $\sigma(H)$; the gap $\Delta = E_1 - E_0$ and its thermodynamic limit are framework primitives, anchored on the certified master hierarchy $\Delta_{\text{DL}} \leq \Delta_{\text{Knabe}}^{(n)} \leq \Delta_{\infty} \leq \epsilon_n$.

S3 (Symmetry and moment map).

A symmetry group G of H with moment map F_{MM} defines, in $\mathbb{CP}(\mathcal{H})$, the constraint manifold $\mathcal{M}_{\text{constraint}}$ with $F_{\text{MM}}(\psi) = 0$ on it — Type-A [=] by the moment-map construction.

S4 (Variational principle).

The ground energy is $E_0 = \inf_{\psi} \langle \psi | H | \psi \rangle$, and the geometric burden $B_{\text{geom}} = \text{Var}(g_{\text{FS}})$ is bounded below by zero — which supplies the required variational structure.

S5 (discharged).

Metric-flow well-posedness was a primitive in v6.0-F and is derived in v6.5-RF8 from the Hessian identity below; the primitive count is four.

S5' (primitive-vs-emergent, structural).

Every physical observable in RCF is emergent from the relational coarse-graining $C_L : B(\mathcal{H}) \rightarrow B(\mathcal{H})/\sim$, never primitive. A substrate’s primitive-observable algebra $\mathcal{A}_{\text{prim}}$ is admissible as a host for the projection picture only if it admits an emergent interpretation under some C_L ; the distance $|\mathcal{A}_{\text{prim}} - \mathcal{A}_{\text{emergent}}(C_L)|$ is the substrate’s primitive–emergent gap.

The frontier programme gave S5' its first positive execution. The AKLT arc had read the axiom as a *ruling-out* instrument (substrates whose primitives are too distant are excluded); the frontier programme read it as a *construction* instrument: the object the axiom demands — an observable that exists only as an emergent of a relational structure — was built, measured, decomposed, and certified natively at machine precision. The census class F4 (Section 4.3) is the axiom’s instrument form.

2.2 The Hessian identity and the robustness principle

Theorem 2.1 (Hessian identity; Type-A [=] (VN-19, established v6.5-RF8)). *For the Fubini-Study metric on $\mathbb{CP}(\mathcal{H})$,*

$$g_{\text{FS}} = \frac{1}{2} \text{Hess}(\Phi), \quad \Phi = 2 \log \langle \psi | \psi \rangle. \quad (1)$$

Proof. In holomorphic coordinates $[z_0 : \dots : z_n]$ the Kähler potential is $K_{\text{FS}} = \log \sum_k |z_k|^2$; the Kähler metric is $g_{i\bar{j}} = \partial_i \partial_{\bar{j}} K_{\text{FS}}$. Setting $\Phi = 2K_{\text{FS}}$ gives $\text{Hess}(\Phi)_{i\bar{j}} = 2 \partial_i \partial_{\bar{j}} K_{\text{FS}} = 2g_{i\bar{j}}$. $\square$

The identity is a tautology of Kähler geometry; the framework’s content is the *choice* of $\Phi = 2 \log \langle \psi | \psi \rangle$ as the canonical potential of quantum state space. Its structural consequences — the well-posedness of the metric flow (the discharged S5), and the hierarchical RP principle — are unchanged. The principle decomposes robustness into two sequential layers: a Type-A instantaneous layer ($d_t F_{\text{MM}} = -L_F F_{\text{MM}} \rightarrow 0$, rate fixed by the symmetry-algebra spectrum) and a Type-B mechanism layer ($d_t B_{\text{geom}} = -\|\nabla B_{\text{geom}}\|_{g_{\text{FS}}}^2 \rightarrow 0$, rate fixed by the Hessian spectrum), with the detectability contraction $\|\text{error}_k\|^2 \sim \gamma_{\text{DL}}^{2k}$, $\gamma_{\text{DL}} = 1/\sqrt{3}$ (VN-17). The frontier programme does not touch this layer; it supplies, for the first time, a *measured instance* of the principle’s deepest commitment — structure that exists only by virtue of a relational act (χ , Section 8) — and an original conservation law at the interface (Section 9.3).

2.3 The epistemic classification

Every vanishing claim in the register carries a tag [?]: **Type-A** [=] — an exact identity forced a priori by symmetry, algebra, or representation theory, constraining the *form* of related quantities; or **Type-B** [$\rightarrow$ rate] — an asymptotic limit whose rate is determined a posteriori by mechanism. The classification’s anti-conflation rules (bridge requirement, mandatory tagging, form-not-rate, default-to-B) govern every claim in this paper. The frontier programme added a third grade family specific to measurement claims — certification grades (Section 4.1) — without altering the Type-A/Type-B spine. Two frontier results are usefully read as endpoint instances of the classification: the response torus’s unit-modulus law (“no amplitude dial”) is a Type-A statement promoted *from* a Type-B absence measurement (Section 9.2); and the summed-exponent law entered as Type-B (measured on χ) and now holds with a falsifier on every future member of its class — the register’s standard route from measurement to law.

2.4 The six-round AKLT arc and the substrate change

The AKLT arc (rounds delta, epsilon, zeta, eta, eta-extreme, Option A; [?, ?, ?]) attempted a concrete realization of the projection picture on the AKLT spin-1 chain under two burden dials ($V_{\text{strain}} = +h \sum_j (S_j^z)^2$ driving the Haldane–large-D transition; $V_{\text{FM}} = -J \sum_j \mathbf{S}_j \mathbf{S}_{j+1}$ driving the Haldane–Ising transition). The arc’s verdict was negative on every dynamical form the substrate could host, and the verdict is permanent scientific capital: a falsified collapse ordering; a falsified role-transfer; a characterized conformal floor artifact (the critical L_0 “floor” is the $v\pi/N$ finite-size regularization of the Haldane–large-D critical point, ratio 0.838 against the $1/N$ prediction at distance 0.005); a characterized multiplet artifact (Lanczos within-multiplet splitting; the corrected singlet-channel gap 0.001231 at $J = 0.3$); a complete map of the substrate’s trade-off surface (gap versus information, monotone, no interior extremum, all channels cloaking together at $h \rightarrow \infty$); and the SPT-edge fragility result of Option A (edge splitting follows the bulk gap’s scaling class in lockstep; the edge has no scaling class of its own).

The structural source of the negative verdict was the primitive–emergent mismatch: the substrate hands over spin operators and their correlations as primitives, while the framework requires observables emergent from a relational coarse-graining. Version 7.0 therefore launched the substrate-change fork with a three-condition criterion (primitive–emergent admissibility; nontrivial boundary–bulk decomposition; kinematic rather than SPT-dynamical). The frontier

programme is the fork's first branch: it kept the spin-1 substrate but changed the tier under interrogation — from the substrate's phases to the *interface between its two books and its two dynamical channels* — and in doing so found, on the same substrate that falsified the projection picture, an interface that generates observables. The lesson is registered as the fork's amendment: the primitive-emergent gap is a property of the *question asked of the substrate*, not of the substrate alone.

3 The frontier substrate and the two-channel model

This section defines the model. Every quantity measured in the programme is defined here, at operator level, together with its dial dependence and its certified window. The definitions are the executable ones: the artifact chain (Appendix C) regenerates every number in this paper from them.

3.1 The chain, the two books, and the kernel

Definition 3.1 (Substrate). The substrate is a spin-1 chain of N sites, $\mathcal{H}_N = (\mathbb{C}^3)^{\otimes N}$, $\dim \mathcal{H}_N = 3^N$, with open boundary conditions. The certified window is $N \in \{6, 8, 10, 12, 14\}$ (statics ladder), $N = 6$ for the joint dynamical cells (the joint-instrument scope declared at R44 and extended by the N-side rungs as certified).

Definition 3.2 (The two books). Two commuting-in-book observables partition the substrate's content:

$$\hat{N} = \sum_{i=0}^{N-1} (S_i^z)^2 \quad (\text{counting book: } m \neq 0 \text{ sites}), \quad (2)$$

$$\mathfrak{M} = \sum_{b=0}^{N-2} P_b^{(2)} \quad (\text{relational book: } S = 2 \text{ bonds}), \quad (3)$$

where $P_b^{(2)}$ projects bond $b = (i, i+1)$ onto the total-spin $S = 2$ sector of $1 \otimes 1 = 0 \oplus 1 \oplus 2$. Both books conserve total S^z ; their spectra are non-negative integers; and their commutator has the certified scale

$$\max |[\mathfrak{M}, \hat{N}]| = \frac{2}{3} \quad (\text{the fifth } \frac{2}{3}; \alpha\text{-invariant, all } N) \quad (4)$$

— the exchange's enabling condition, machine-exact at $N = 6, 8, 10, 12$.

Definition 3.3 (The kernel and the two vacua). The *relational vacuum* is the kernel of $\mathfrak{M}$: $\varphi_0 \in \ker \mathfrak{M}$, $\dim \ker \mathfrak{M} = 4 = 2^2$ (the valence-bond construction: two virtual edge doublets, deterministic, edge-mode-explicit). The *counting vacuum* is the product state $|0\rangle_C = (|m=0\rangle)^{\otimes N}$. Their book contents are certified exactly:

$$\begin{aligned} \langle \hat{N} \rangle_{\varphi_0} &= \frac{2N}{3} \quad (\text{site RDM} = \mathbb{I}/3 \text{ exactly}), & \langle \mathfrak{M} \rangle_{\varphi_0} &= 0, \\ \langle \hat{N} \rangle_{|0\rangle_C} &= 0, & \langle \mathfrak{M} \rangle_{|0\rangle_C} &= \frac{2(N-1)}{3}. \end{aligned} \quad (5)$$

The kernel's counting profile $\langle \hat{N} \rangle$ over $\ker \mathfrak{M}$ spans the *band* $[N_{k,\min}, N_{k,\max}]$ around $2N/3$ with spread closing exponentially in N (law ??, Section 5.3); the canonical φ_0 sits exactly at $2N/3$ at every N . The mid-chain entropy of the kernel states is $S_{\text{mid}} = \ln 2$ exactly (the information floor), with edge-singlet composites at $\ln 4$.

The constant $c = 2/3$ is the substrate's kinematic signature, and the register holds it as a *family*, not a coincidence: the per-bond kernel content $s = 2/3$; $\langle \hat{N} \rangle_{\varphi_0} = 2N/3$; the commutator scale, Eq. (4); the dynamic landing $B_\infty = c$ (the Jordan channel fills the counting book to exactly the kernel's equilibrium content, R40, machine-exact to 0.05%); and the kernel-profile deviations closing exponentially. One kinematic family, one dynamic landing — the c -suspicion was closed at R41 and has not reopened since.

3.2 The two channels

Definition 3.4 (Lie channel (transport)). The *Lie channel* is the unitary spin-1 XY dynamics

$$H = \sum_{j=0}^{N-2} \left(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y \right), \quad J = 1, \quad (6)$$

which conserves $\sum_j S_j^z$. It is the substrate's *transport* class: ballistic fronts (free-sector velocity $v = 1.6444 \pm 0.0276$ against $v_{\max} = 2 \min(J, g) = 1.5868$ at the stationary point $k^* = \arccos(g/J)$), and the sole carrier of counting–book interconversion at early times.

Definition 3.5 (Jordan channel (reconciliation)). The *Jordan channel* is the bond-local dissipative damping of the $S = 2$ bond population, with Clebsch–Gordan-declared jump pair on each bond b :

$$L_1 : |2, M\rangle \rightarrow |1, M\rangle \quad (M = -1, 0, 1), \quad L_2 : |2, \pm 2\rangle \rightarrow |1, \pm 1\rangle, \quad (7)$$

at rate dial γ , satisfying $\sum L^\dagger L = \gamma P^{(2)}$ *exactly* (the $S_0 + S_1 + S_2 = \mathbb{I}_9$ completeness of the bond algebra after the A42-1 repair, residual 2.8×10^{-16}). It is the substrate's *reconciliation* class: it flows the counting vacuum into the kernel while filling the counting book to exactly the kernel's equilibrium content ($B_\infty = 2/3$; multi-stage decay $-4.22 \rightarrow \lambda_{\text{eff}} 0.065 \rightarrow \lambda_{\text{tail}} 0.061$; 99.24% decay at $t = 60$; $K_\infty = 0.971$ rising; $|M_z| \leq 8.8 \times 10^{-17}$ throughout).

The two channels are complementary by construction (R39, machine-confirmed in both sectors): transport is Lie-indexed, reconciliation is Jordan-indexed, and the channel split is the substrate's first certified structure. Their composition classes are the two orderings of the joint evolution, below.

3.3 The crossing: joint dynamics and its two orderings

Definition 3.6 (The crossing). The *crossing* is the joint Liouvillian

$$\mathcal{L} = -i[H, \cdot] + \sum_b \sum_{\mu=1,2} \left(L_{b,\mu} \cdot L_{b,\mu}^\dagger - \frac{1}{2} \{L_{b,\mu}^\dagger L_{b,\mu}, \cdot\} \right), \quad (8)$$

acting on density matrices $\rho(t) = e^{\mathcal{L}t} \rho(0)$ (Lindblad form [?, ?]; integrators cross-validated expm versus 40th-order Taylor to digit agreement, traceA cross-validation 2.8×10^{-15}). The *two orderings* of the crossing are the strobed cycle maps at fixed total exposure, τ per cycle, split s :

$$C_{LJ} = e^{(1-s)\tau\mathcal{L}_J} e^{s\tau\mathcal{L}_H}, \quad C_{JL} = e^{s\tau\mathcal{L}_H} e^{(1-s)\tau\mathcal{L}_J}, \quad (9)$$

with $s = 0.5$ (equal exposures) and $\mathcal{L}_H = -i[H, \cdot]$, $\mathcal{L}_J$ the dissipator of Eq. (7). Their periodic steady states are the PERM instrument's cells (Section ??); the order separation between them is the substrate's *order memory*.

Remark 3.7 (The factorised tier). The *upstream* (un-crossed) tier is the product dynamics: the Lie channel alone (e^{-iHt}), the Jordan channel alone ($e^{\mathcal{L}_J t}$), and their statistically independent composition. Upstream the channels are *factorised by definition* — there is no joint measure, only the product of the two channel measures. This is the structural fact that the nativity test (Section 8) exploits: the crossing's act is the construction of the joint measure, and its connected residue is χ .

3.4 Dials, windows, and the two frontiers

α (composite dial).

The composite operator $\mathfrak{M}(\alpha) = \mathfrak{M} + \alpha \hat{N}$; ground states $E_0(\alpha; N)$ define the static anatomy. Certified grid $\alpha \in [0.05, 0.30]$ at every rung.

γ (channel-rate dial).

The Jordan rate; certified window $\gamma \in [0.1, 500]$ (scan anchor $\gamma = \mu_1 = 0.413240$, the free-sector gap $2(J - g)$). The two-zeros tail runs $\gamma \in \{30, 100, 300, 500\}$.

 N (size dial).

6, 8, 10, 12, 14; 14 is the sealed-rule rung (Section ??).

 τ (clock).

Evolution time; the memory instrument runs $\tau \in \{0.125, 0.25, 0.5\}$ per cycle with saturation windows declared per cell class.

 Γ (substrate coupling).

The chart dial of Section 10, with μ_1 the principal book value; the un-crossed exclusion boundary is certified linear in (Γ, μ_1^2) .

Definition 3.8 (The two frontiers and the margin). The α -frontier (static) is the composite envelope $\text{ENV} - \text{int}(N)$: the PCHIP interpolant of the certified anatomy rows in (N, M) book space, with the kernel face $M = 0$ on $[N_{k,\min}, N_{k,\max}]$ — a strictly convex curve meeting the kernel band at the relational corner. The γ -frontier (dynamic) is the NESS occupancy curve: the book-space locus of the non-equilibrium steady states $\bar{\rho}^*(\gamma)$ of the crossed dynamics, certified by its seven-point scan (all *burdened*-NESS; escape margins $\Delta_{\text{env}} = 1.1603, 1.1582, 1.1523, 1.0962, 0.9005, 0.5835, 0.1935$ at $\gamma = 0.1, 0.3, 0.41324, 1, 3, 10, 30$ respectively — strictly monotone, no crossing). The two are *distinct curves* meeting only asymptotically at the relational corner — R42’s certified two-frontier structure. The *margin field* is the escape of the joint dynamics above the static frontier:

$$\Delta(t) = M(t) - M_{\text{env}}(N(t)), \quad (10)$$

measured on trajectories from either vacuum. The margin is the programme’s central observable: everything after R42 is either its decomposition, its zeros, its nativity, or its return path.

The NESS scan itself carries the R39–R40 capacity structure: the NESS at declared rates fails on both sectors (free-sector asymmetry $18.69\times$; interacting, replicated), all operating points are burdened, $\gamma^* > 30$ (balance extrapolation ~ 35), and n^* climbs $0.595 \rightarrow 0.662$ toward $c = 2/3$ as γ grows — the equilibrium burden profile is the Jordan channel’s attractor, dynamically selected rather than chosen.

4 The verification apparatus

The frontier programme’s results are only as strong as the instrument discipline that produced them, and that discipline is itself part of the framework — the register treats the apparatus as infrastructure with its own certified invariants. This section specifies it.

4.1 Certification grades

Every number in this paper carries a declared grade: *machine-identity* ($\leq 10^{-12}$ against a certified counterpart; the bit-exact class is stronger still — byte-level reproduction across fresh re-executions); *statistical* (a fit with declared rmse, window, and family); *absence-grade* (a verified null at a declared resolution floor); and *citation* (a value certified in a prior round and imported by hash, not re-run). Windows are declared per cell class (headroom fields on every sheet since F-V4-5); tolerances are chosen, not inherited (the V7-D2 lesson); and every quoted value regenerates under a *named extractor* (the quote-with-convention rule, V-4).

4.2 Sealed predictions and pre-commitment

Round sheets are pre-registered before execution (the print order is the seal), keyed by a sheet hash under which all checkpoint cells are stored; imports cite the generating sheets' hashes and the generator scripts' md5, byte-compared at import. Quantitative predictions are *sealed* — text, hash, and timestamp bound before the data exist — and the seal is one-way: any post-hoc edit voids the dependent rung. The prediction ledger holds ten sealed predictions (A–J; Appendix B): eight confirmed, one killed at sealed scope (F, Section 10.3), two pending (I, J, Section 13). The register's own honesty note is part of the record: an unbroken confirmation record is evidence that questions were chosen where instruments could already constrain answers, not that the seals are soft — and the discipline's real test, a kill, arrived at R49 and was taken without re-grading or resurrection.

The pre-commitment machinery's institutional signature is the *rule closure*: at R45 the register's second standing question (Kill #2, the macro-limit collapse hypothesis) was closed not by a threshold crossing but by a sealed adjudication rule — text hash-chained before the first $N=14$ cell executed, stating the verdict conditions, executed mechanically on the rung's reading (Section ??), and audited post-use with the hash chain proving the text never changed. Two closure modes now sit in the archive side by side: closure by rule and closure by falsifier.

4.3 The crossover-difference census

The census is the axiom S5' in instrument form, executed at V-8 and maintained since. For every registered observable X with a certified face on both sides of the crossing, the census classifies $\Delta_F(X)$ — what crossing the frontier does to it:

Class	Name	Falsifier
F0	identity preserved	any certified-scope discrepancy
F1	value shift	— (mechanism column required)
F2	behaviour shift (law rewrite)	law equivalence under refit
F3	identity split ($1 \rightarrow n$)	member re-merge under any probe
F4	emergent observable	derivation from upstream alone (the fingerprint test)
F5	combination ($n \rightarrow 1$)	additivity null, absence-grade certificate required for demotion
F6	annihilation (no downstream face)	any surviving downstream reading

Two discipline rules govern the table: every class with a one-directional falsifier must bind a null at seal time (verifier symmetry, the V8-D2 rule), and rows enumerate by *certified scope*, not by name (the V9-D1 rule, globalized). The census's credential is that it overrode its own charter's candidate labels twice in its first execution (margin drafted F1, executed F2; curvature drafted F1, executed F3) and discriminated in both directions in the nativity test (native for χ , derived for the approach exponent). Its final state — twenty-three rows, distribution six held, three shifted, one rewritten, three split, five born, two buried, one joined (two faces), two unpairable — is Table 2.

4.4 The defect taxonomy

The verification interludes name their own defects, in the round, before adjudication. Across ten consecutive rounds the defects fell into exactly three families, with zero new families:

1. **Incomplete enumeration** (V6-D1, V7-D1, V8-D1, V9-D1, V10-D1, V11-D1, V12-D1, V13-D1): a first-pass specification enumerates by name and misses an object whose faces carry different names. Structural fix: scope-keyed enumeration, globalized at V-9. The family's cost curve descends to zero (pre-execution catches from V-10 onward) — the repair-propagation

invariant *a structural repair is local until explicitly generalized* received its first confirmed save at V-10 and its third at V-13.

2. **Inherited assumption** (V7-D2 through V13-D2): an instrument inherits a choice one layer deeper than it was chosen for. The family’s descent through the instrumentation stack is itself a registered invariant: *phrase* (V-6) $\rightarrow$ *tolerance* (V-7) $\rightarrow$ *verifier symmetry* (V-8) $\rightarrow$ *window* (V-9) $\rightarrow$ *parameter family* (V-10) $\rightarrow$ *statistic* (V-11) $\rightarrow$ *grid* (V-12) $\rightarrow$ *basis* (V-13). The register’s standing reading: the descent is the apparatus’s regeneration cost, with asymptote zero marginal cost, not zero; and at the basis layer the auditor’s correction and the physics discovery were the same event (the torus, Section 9.2) — the descent bottoms out exactly where the apparatus’s inheritance and the physics’s representation become the same choice.
3. **Budget window** (V6-D3 through V13-D3): a cell exceeds its compute window; killed at the edge, re-routed to cite-don’t-rerun with provenance flagged. The structural ritual — the one place per round where the budget discipline visibly bites.

The taxonomy’s stability is infrastructure behaviour, and it is load-bearing for this paper: an apparatus whose failures are predictable in class and unpredictable only in surface is one whose confirmations mean something. The interlude-by-interlude record, with cell counts, is Appendix C.

5 The static frontier

5.1 The composite anatomy and the deficit law

The static instrument diagonalizes the composite $\mathfrak{M}(\alpha) = \mathfrak{M} + \alpha \hat{N}$ (the $S^z = 0$ block is exact by symmetry: both books conserve S^z and the composite ground states live in $S^z = 0$; the block spectrum equals the full spectrum to 1.1×10^{-14} at shared rows). The composite ground envelope shows *no crossover*: every $\alpha > 0$ grid row is intermediate (the kernel holds the ground only at $\alpha = 0$ exactly), and both pre-registered critical- α candidates are dead. The parabolic corner-cutting is quantified by the deficit law — the excess of the counting line over the composite ground:

$$D(\alpha; N) = \alpha N_{k,\min}(N) - E_0(\alpha; N) = C(N) \alpha^{\beta(N)}, \quad (11)$$

fitted on the pre-registered grid $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ at every rung (extractor KAPPA-c, declared before first execution under the F-V4-2 binding condition; the dead anchors 0.14, 0.117, 0.106 of the pre-ladder era are not reused). The certified ladder: $\beta = 1.9758, 1.9736, 1.9727, 1.9509, 1.951$ at $N = 6, 8, 10, 12, 14$ respectively. The law is *not* an exact two-exponent: the weak confirmation L-44 (clause iii failed at $\sim 2\sigma$ on the last rung) registers the deficit power’s own N-structure — the same finite-size softness class as the curvature law’s drift, below.

$D(\alpha; N) = C(N) \alpha^{\beta(N)}$ with β drifting toward 1.95 across $N = 6$ –14; fitted per rung at rmse ≤ 0.0098 ; falsified as an exact α^2 at the last rung. The drift is itself registered: the parabolic corner-cutting is N-soft.

5.2 The curvature law and the mesoscopic corner

The curvature of the composite frontier at the relational corner — the Legendre dual of the deficit law — is $\kappa(N) = 1/(2C(N))$: near the corner, $M_{\text{env}}(N') \sim (N_{k,\min} - N')^2/(4C)$. The certified ladder:

N	6	8	10	12	14
$\kappa(N)$	0.2938	0.2497	0.2138	0.1932	0.1754
$\beta(N)$	1.9758	1.9736	1.9727	1.9509	1.951

$\kappa(N) \propto N^{-\gamma_C}$ with $\gamma_C = 0.6131 \pm 0.0187$ (four-point) and 0.6147 ± 0.0161 (five-point refit at $N = 14$; the rung lands 0.23% below the four-point extrapolation). The corner's *class* is mesoscopic: γ_C lies two standard errors from both the edge class (0) and the bulk class (1). Corner width $\sim N^{0.31}$ in total book units — a finite, well-defined mesoscale object. The law's own drift (+0.0016 per two rungs) is registered: the corner is N-soft, not N-hard.

The mesoscopic class verdict is the frontier's first spatial-class certificate, and it is what the programme's later rounds lean on: a genuinely intermediate object, with its own width exponent, its own decay exponent, and (Section 11) its own unitary failure mode.

5.3 The band-closing law and the non-echo certificate

The kernel band closes exponentially — the counting spread of $\ker \mathfrak{M}$ around $2N/3$ decays as $\varepsilon(N) = 2N/3 - N_{k,\min}(N) \sim e^{-N/\xi}$ with $\xi \approx 1.05$ sites (closing ratios 0.148/0.139/0.133, mildly accelerating; the local scale at $N = 14$ is 1.026, the band echo 1.55×10^{-6} , the extractor's machine horizon at projected $N \approx 19 \pm 2$). The static projection test then separates the two objects decisively:

$$\kappa/\varepsilon = 26.7 \rightarrow 153.6 \rightarrow 947.0 \rightarrow 6417.2 \quad (N = 6 \rightarrow 12; \times 240 \text{ across the ladder}). \quad (12)$$

The curvature is *not* the band edge's echo: the band closes exponentially, the curvature decays as a power — two objects, two N-laws. The static-projection fork (frontier as the books' projection at the corner versus frontier as independent object) adjudicates on the independent-object side, and the later rounds (R46–R49) convert this separation into composition: what the projection lacks is precisely the interference carrier (Section 7).

5.4 The three-way zero ordering

The decomposition-completeness question — does transport account for all of the margin? — was sealed as a fork with pre-committed families and a declared resolution floor $\varepsilon_0 = 10^{-8}$. On the γ tail $\{30, 100, 300, 500\}$:

γ	30	100	300	500
margin Δ	0.1935	0.0243	0.00287	0.00104
drive T	0.2741	0.1047	0.0364	0.0220
$\Delta/ T $	0.706	0.232	0.0788	0.0475

POWER fits win decisively over stretched-exponential and contact alternatives (rmse 0.029/0.021 versus 0.31/0.13 and 0.27/0.12): $\Delta \sim 114.8 \gamma^{-1.859}$, $T \sim 6.10 \gamma^{-0.899}$, ratio exponent -0.9606 ± 0.0166 (-0.96 at 2σ from zero). Floor crossings at $\varepsilon_0 = 10^{-8}$: $\gamma_\Delta = 2.6 \times 10^5$, $\gamma_T = 5.9 \times 10^9$, with the projected zero bracketed $\gamma_{\text{proj}} \in (30, 100)$:

$$\gamma_{\text{proj}} < \gamma_\Delta < \gamma_T, \quad (13)$$

robustly separated across three orders of magnitude. The margin dies roughly twice as fast as its transport driver; the decomposition is **incomplete** — **non-transport carriers exist**. (The ordering’s composition-level attribution — γ_Δ is χ ’s floor crossing — arrives at R46 and is stated with law 9.)

5.5 Transport saturation: the N-side half of the same verdict

The transport-only margin family (Lie channel alone, from φ_0) saturates per bond:

$$\Delta_T(t=5)/(N-1) = 0.0529 \rightarrow 0.1500 \rightarrow 0.1489 \rightarrow 0.1468 \rightarrow 0.147$$

across the ladder — the $N = 6$ cell is finite-size, and the per-bond transport margin is a *bulk object* with a flat, N-independent profile. Consequence: the *N-side transport zero does not exist*. Together with the γ -side ordering, the zero structure closes symmetrically: on the γ axis the margin dies at twice the transport rate (non-transport content exists); on the N axis transport saturates while the margin itself turns (Section 6.3) — the extrapolated zeros of the pre-census era are projection artifacts, and the margin is a non-transport object with (after R46) measured composition.

6 The dynamic frontier

6.1 The two-frontier separation

R42 certified the dynamic frontier as a tier of its own. All seven NESS points sit strictly off the static frontier (Δ_{env} strictly monotone in γ , no crossing, clause margins 4–23 $\times$ the floor), and the lean L-42 (“frontier-maker”, declared with bias flag on) was **confirmed-strong** at 7/7. The two frontiers — the α -frontier of the algebra tier and the γ -frontier of the joint tier — are distinct curves in book space, meeting only asymptotically at the relational corner. The escape-margin field between them is the first measured content of “the substrate contributes”: the margin of Eq. (10).

6.2 Channel typing

The sharpened question of R42 — corner-mover or frontier-maker? — resolved as a *channel-typed split*, and the split itself is the datum:

Channel	Typing	Certified reading
Jordan (from φ_0)	corner-keeper	frozen on the kernel face at machine zero ($d_{\text{env}} = -4.2 \times 10^{-17}$; all X-rates void)
Jordan (from $ 0\rangle_C$)	corner-selector	attractor ON the kernel face (landing +0.0253 at $T = 60$)
Joint	frontier-maker	attractor +1.10 off the static frontier at the operating point; kernel-band crossing at $t = 1$ (the minting overshoot datum)
Lie (from φ_0)	slope-breaker	realized -1.03 in transit versus the slope field -0.24
Lie (from $ 0\rangle_C$)	on-slope trader	initial trade rides the frontier’s local slope (+0.003 mismatch at $t = 0.5$), then departs

The freeze at the relational corner is certified static as well: the G3 void at 10^{-16} — the relational vacuum is the Jordan channel’s fixed point. Front-level structure is similarly typed: the two books’ edge–mid gaps disagree at $t = 0.5$ and $t = 2$, agree at $t = 0.25, 1, 5, 20$ (the observable-class SPLIT cell), and the T' -arrival table types the orderings: Lie flat; joint edge-lagged by 0.300; Jordan edge-lagged by 0.500 — amplitude overshoot without front lead.

6.3 The margin’s persistence and the N-side turn

The margin’s N-side ladder was the kill #2 instrument. The transport-accounting ratio $X(N) = |X-t_1(N)|$ reads 1.0306, 1.0179, 1.0223, 1.0294, 1.0302 at $N = 6, 8, 10, 12, 14$: the rung step $-0.0012 \rightarrow +0.0008$ *reverses* at the sealed-rule rung, independently confirmed by the X-map trajectory (inflection in-trajectory, both instruments agreeing to 0.4 rung).

The pre-registered firing clause was mechanical: not-fired iff net decay and second-half envelope strictly inside first-half. The rung survived its window by 0.0012 (4%) at $N = 12$ with a rising tail; the $N=14$ rule was sealed before the first cell ran (*the margin either doubles the gap or reverses; under the standing table neither survival nor threshold crossing occurs*), and the rung read the reversal branch. Kill #2 is closed — not fired, ruled — the register’s first question closed by pre-commitment rather than by threshold or adjudication. The “drift vanishes with N” clause is mechanically dead on both sides: transport side (flat per-bond profile, Section 5.5), net side (tail reversal).

The reversal is the margin’s *persistence* certificate, and its arithmetic is the round’s structural finding: post-turn, the margin grows with the corner. The post-turn slope $+0.0008$ per rung is 55% of the corner-width growth rate (4.9% per rung). The two axis-profiles of the same object are sign-opposed — on γ it dies at twice transport’s rate, on N it turns and grows — and a projection artifact has no mechanism for dying on one axis while growing on the other. The N-side turn is the third axis of the projection-only reading’s failure.

6.4 The 55% lever

The margin’s growth rate equals the corner’s growth rate times the corner’s mass fraction:

$$\begin{aligned} \text{slope}_\Delta &= (C/\Delta) \times \text{slope}_w = 0.551 \times 4.9\% \\ &= 2.70\%/\text{rung} = +0.0008/\text{rung}, \end{aligned}$$

exact to display precision, and stress-covariant (Section 11). The inter-layer coupling number and the composition number are one number: the static frontier’s corner carries the dynamic frontier’s margin by mass fraction, through mixture.

This is the framework’s first measured inter-tier coupling: the layer above reaches into the layer below by exactly its share of the mass. The identity’s certification history is the programme in miniature — measured as an unexplained coincidence at R45, closed as mechanism at R46 (the non-corner carriers are N-flat; only C tracks the corner), stress-tested at R50.

7 Fusion: the carrier decomposition

7.1 Instruments and the decomposition

Two new static-side instruments executed at R46, both with certified purity: **excision** (the corner’s contribution removed by the R43-certified corner mask; zero-excision returns baseline

byte-exact) and **blocking** (channel masks; zero-block is the identity; round-trips exact). The over-determined closure of the block ladder against the excision cross gives the decomposition at the operating point ($N = 14$):

$$\Delta = C + L + J + \chi = 0.0166 + 0.0047 + 0.0027 + 0.0062 = 0.0302, \quad (14)$$

with C the Jordan corner-keeping carrier (the two instruments cross at $C = 0.0166$ to 0.03%), L and J the Lie-first and Jordan-first channel contributions, and χ the residual. The corner's mass fraction is $C/\Delta = 0.551$ — the lever of Eq. (??), now with a numerator.

7.2 The F5 verdict: the frontier combines

The sealed additivity null — $\Delta \stackrel{?}{=} C + L + J$ — reads 0.0240 against the measured 0.0302: the residual $\chi = +0.0062$ is 20.5% relative, two full orders above the absence threshold. The absence-grade certificate is correctly *not* issued:

The LJ/JL channel contributions fuse at the crossing; the net exchange field is a genuine, constructive combination. The census's routed row resolves to F5 with its mechanism certified: the transformation of the exchange rate (slope $\rightarrow$ field) *is* this fusion — the field is the interference sum of the two slopes. The margin's census entry re-registers dual-label, F3 + F5, both reading directions named (upstream: combine; downstream: split into four certified carriers).

The frontier's transfer rule — the census's distribution read as a verb — gained its first positive combine-axis data point at R46 after V-8 had registered the asymmetry (splitter, not joiner) with the honest caveat that the combine null was untested. The junction is also *generative*: census row 22 (χ itself) registers F5-born, no upstream standalone face — the first object whose nativity is constitutive (Section 8).

7.3 The mixture mechanism of the turn

The decomposition explains the N-side turn mechanically. Pre-turn, the non-corner carriers decay ($0.0155 \rightarrow 0.0136$ over $N = 10 \rightarrow 12$); post-turn they are N-flat (0.0136 at both $N = 12$ and 14 , flat to 0.3% — the same per-bond saturation class as transport's 0.147 profile), while C tracks the corner width exactly. The composite slope's sign flipped when the saturating component stopped overwhelming the growing one: the margin did not change behaviour at $N = 14$; its *mixture* did. This retroactively grounds the V-8 census override — margin drafted F1, executed F2 — at the root: the composite law rewrites because it is a fusion of two simple laws. F2 was the shadow of an F5.

7.4 The two-component profile and the exponent's owner

Decomposed on the γ dial, the margin's celebrated $\gamma^{-1.859}$ splits:

$$\chi \sim \gamma^{-1.861}, \quad (C + L + J) \sim \gamma^{-0.518}, \quad (15)$$

two-component refit of the original profile: rmse $0.029 \rightarrow 0.007$ (fourfold; the R44 residual tension resolves as the fusion’s fingerprint). The “margin dies at twice transport’s rate” belongs to χ alone (decomposed: $2.07\times$ transport, while the substrate dies *slower*, $0.58\times$); $\gamma_\Delta = 2.6 \times 10^5$ is χ ’s floor crossing. The ordering of Eq. (13) acquires attribution: *projected artifact* < *interference-carrier death* < *transport death*. A single exponent fitted to a two-carrier composite is wrong for both carriers in the window where both live — the composite-projection trap, now with a certified instance.

The parents’ individual profiles ($L \sim \gamma^{-1.118}$, $J \sim \gamma^{-0.743}$, each at its own certified window and family) close the arithmetic: the cross-term carries the *summed* decay rate of its parents, which is the subject of law 7 below.

8 The source: constitutive nativity of χ

8.1 Upstream absence on three routes

The nativity test ran three independent routes to the same question — can χ be derived without the crossing? — and all three returned the strongest available form of “no”:

Route	Method	Result
1. Operator algebra	the cross-channel composition as an operator object on the factorised tier	no nonzero object exists: the connected composition is identically the zero operator
2. Correlator	the connected part of the channel cross-correlation upstream	identically zero — factorization is what “channel” <i>means</i> upstream
3. Simulation	un-crossed system under the certified χ extractor	0.00e+00, byte-level

$\chi_{\text{upstream}} = 0$ on three routes; scope: absence certified within the window, impossibility mechanism certified within the tested formalism. The upstream does not merely lack the value of χ — it lacks the *capacity* for χ : the factorised tier has no joint measure, and therefore no well-posed object for any derivation to reach. χ is the census’s first object born by combination and unreachable from any upstream ledger.

8.2 The de-factorisation mechanism

The mechanism is the round’s physics payload, and it is exact in form: *the crossing de-factorises*. Upstream, the channels are statistically independent by construction — the joint state is the product measure. The crossing’s act is the construction of the joint measure; χ is its connected residue:

$$\chi = \langle \mathcal{O}_L \mathcal{O}_J \rangle_\times - \langle \mathcal{O}_L \rangle_\times \langle \mathcal{O}_J \rangle_\times, \quad (16)$$

normalised to the margin metric. The generate-versus-reveal question — is the fusion creating a new object or exposing a hidden old one — resolves as **generative**: there was nothing to reveal. This is the strongest nativity grade the framework’s instruments can issue, and it is the same grade the torus law later received (Section 9.2): measured absence promoted to structural no-slot.

8.3 The order coupling

The sealed prediction (A) — interference implies order-sensitivity — confirmed with the interference signature: under channel transposition ($LJ \leftrightarrow JL$), χ *flips sign* with $|\chi|$ preserved to display precision; the other carriers are order-invariant to first order; χ is the decomposition's only order-responsive member. The response vector correlates with the registered order-memory dial κ_{perm} at 0.93. Three R44 readings retroactively decompose as χ 's fingerprints: the 6.5% steady-state order sensitivity is $2\chi \times f_\tau$ with $f_\tau = 0.158$ (display precision); the JL off-curve residual $+0.0129$ is 2χ (within 4%); and $\kappa_{\text{perm}} \neq 0$ is χ 's order-face. The frontier's order memory is not a separate phenomenon — it is the phase of the same act whose amplitude is the fusion.

8.4 The summed-exponent law

The interference carrier inherits the *summed* decay of its parents:

$$p_\chi = p_L + p_J = (-1.118) + (-0.743) = -1.861, \quad (17)$$

closure at display precision, surviving the family-free log-derivative extraction (1.862, 0.05%). Certified external objects: the corner's stress-response amplitude ($\gamma^{-1.859}$, 0.1% match, R50) and the exclusion chart's order shift ($\gamma^{-1.861}$, 0.2% match, R49) — neither an interference term by construction, both carrying the interference's decay arithmetic. **Falsifier:** any future cross-term measured under the programme's instruments must obey the summed-exponent arithmetic, or the interference mechanism dies. The law is now three-object, machine-held.

8.5 The approach exponent: the discrimination cell

The approach exponent of the NESS scan toward the frontier pair (extracted by RELAX from the upstream relaxation spectrum) matches the JL channel's slow mode to 0.3%: census row 19 resolves unpairable $\rightarrow$ F1, transformed, mechanism named (the crossing's freeze couples the trajectory's approach to the substrate's slowest channel mode). The cell's methodological weight exceeds its physics: the nativity instrument returned *native* for χ and *derived* for the approach exponent in the same execution — the two-direction discrimination that certifies the instrument itself. An instrument that says yes to everything certifies nothing; this one returned both verdicts, and both held.

9 The return path: formation, torus, listening

9.1 Formation kinetics

The order-memory dial's full response $\kappa_{\text{perm}}(N, \gamma, \tau)$ collapses onto a single curve in one variable — the formation law:

$$\kappa_{\text{perm}}(N, \gamma, \tau) = g\left(\frac{\tau}{\tau_{\text{mem}}(N, \gamma)}\right), \quad \tau_{\text{mem}} = \tau_0 N^{0.31} \gamma^{-0.743}, \quad (18)$$

collapse residual $\leq 2.1\%$ across all thirty-six cells ($N \in \{10, 12, 14\}$, $\gamma \in \{1, 2, 3\}$, four clocks). The formation clock is the JL slow mode (τ_{mem} 's γ -exponent matches J 's to 0.4%) scaled by the corner width ($N^{0.31}$, within fit error) — the substrate's slowest channel mode is the frontier's internal time, applied twice: once to the approach (Section 8.5), once to the memory's construction.

Three consequences, each closing a standing item. First, the γ -growth of the raw dial ($\times 2.2$ over $\gamma = 1 \rightarrow 3$) is kinetic, not structural: the dial measures the formation *fraction* ($0.44 \rightarrow 0.97$ development), while the capacity — the saturated amplitude — is $2|\chi|$ and decays by the summed-exponent law. The memory does not grow with γ ; it forms faster. Second, the flow's steady state is not the memory's steady state: the R44 steady-state reading was g at the flow's equilibrium clock, $f_\tau = 0.158$ — the memory outlives the flow's equilibration and is still forming at 15.8% when the flow has settled. The apparent tension was two clocks, never compared; now compared. Third, on the N axis at fixed τ the dial *falls* while the capacity holds (N-flat, $\pm 0.4\%$): the memory is size-slow, not size-small.

9.2 The response torus

The saturated order response over the full enumeration — 639 cells: all 71 non-identity permutations $\times N \in \{10, 12, 14\} \times \gamma \in \{1, 2, 3\}$ at the saturated clock — has binary amplitude: every non-identity permutation returns $|R| = 2|\chi|$ within 0.3%, with zero cells in the hollow core $(0.03, 1.97) \cdot 2\chi$ (the R48 sampling caveat discharged; the R50 full enumeration, seal H). The response's geometry, found in the response's own eigenbasis (the V13-D2 catch, where the auditor's correction and the physics discovery were the same event):

$$R(\sigma) = 2|\chi| \cdot u(\theta_L(\sigma)) \otimes u(\theta_J(\sigma)), \quad |u| = 1.0000 \pm 0.0003 \text{ on all 639 cells.} \quad (19)$$

The response space is a *two-torus* — two phase dials, one per channel — whose projection in the inherited channel basis is the shell the sampling era registered as a sphere (torus residual 0.4% versus sphere 8.1%; dial variances match the LJ and JL phase spectra at 0.91/0.89; permutation map Haar-uniform, KS $p = 0.62$). The identity permutation is the torus's missing point — the one rearrangement with no phases to conjugate. Because $|R| = 2|\chi| |u_L| |u_J|$ with unit-modulus factors, *there is no amplitude dial to turn*: the interior-emptiness law is an arithmetic identity within the tested formalism. The memory has one bit of amplitude that is not a bit at all, and a full vector of direction: it remembers *which way*, never *how much*. The phases are the channels' own — the crossing did not invent the language's letters; it conjugated them.

9.3 The listening law and exponent conservation

The sealed prediction (D) — the kernel's response tracks the interference carrier — confirmed with lag, gain, and upstream absence, all three links independently measured:

$$\delta K(\sigma, \tau) = \Lambda(\gamma) R(\sigma, \tau - \tau_{\text{mem}}), \quad \tau_{\text{lag}} = 1.02 \pm 0.02 \tau_{\text{mem}}, \quad \Lambda \sim \gamma^{+1.861}, \quad (20)$$

with the product $\delta K = +0.0146 \pm 0.0004$ γ -flat across a window in which the source χ decays by a factor 7.7; upstream $\delta K = 0.00\text{e}+00$ (byte-level: before the crossing, channel order does not touch the kernel at all). The kernel hears on the corner's clock, at cancellation gain, deaf to amplitude, tuned to direction (correlation 0.96 at lag; the zero-lag statistic 0.71 was the wrong statistic — the V11-D2 lesson, statistic choice as a seal-time declaration). **Exponent conservation:** χ decays at the summed rate of its parents; Λ grows at exactly that rate; the product is scale-free:

$$p_\chi + p_\Lambda = 0. \quad (21)$$

The crossing creates the interference at summed decay and the back-coupling at summed growth, and the interface conserves the exponent across both laws — the framework's first original conservation law (the earlier arithmetic identities were product-with-factors; this one is a conservation of a process across the act that creates the interface). **Falsifier:** any future back-coupled channel must show the same cancellation, and the γ -flatness must survive window extension, or the law dies. Certified within the window (flat across a factor-8 source decay); the R43 extrapolation caution applies beyond it.

The back-coupling is phase-coded: amplitude-invariant, direction-tracking. The kernel's response is addressed by exactly what χ alone carries — the sign, the arrangement, the memory — and the census's K-response row re-registers as the register's first entry with *certified nativity and upstream address*: born at the crossing, delivered to the substrate. The causal chain $\text{crossing} \rightarrow \chi \rightarrow K$ is the programme's first fully measured loop segment, and it converts the founding question one final time: the interface is not merely a source — it is a channel back into the substrate that built it.

10 The inscription: the exclusion chart

10.1 The palimpsest

The exclusion chart — the boundary of the admissible regime in the (Γ, μ_1) plane, drawn crossed and un-crossed — is a *palimpsest*: two authors, one localized overlap. Un-crossed, the boundary is smooth and straight in (Γ, μ_1^2) :

$$\mu_1^2 = a\Gamma + b, \quad \text{linearity certified to } 3 \times 10^{-4}, \quad (22)$$

the algebra's signature on the substrate's own document, order-invariant to absence grade. Crossed, the same line carries a localized bulge: the band's Γ -interval matches the corner's projection (R43 geometry cited), its width matches the corner width, its amplitude profile is the corner mask's shape. The band is present at identity permutation — χ 's baseline face — and it moves under transposition.

Neither sealed branch survived: the chart is neither substrate-autonomous (F's implicit global claim) nor frontier-permeated (the diffuse reading). It is **local inscription**: the substrate's document carries the joint tier's writing exactly where the joint tier lives — the jointness is local, and the overlap has the geometry of the thing that wrote it.

10.2 Prediction E: the payload

Under transposition the boundary shifts by 1.72% ($N = 14$; 1.68% at $N = 12$ — N-flat, χ -consistent), with the shift's sign pattern correlating 0.94 with the χ phase vector on the boundary and round-trip exact. The payload's anatomy:

1. **Addressed, not broadcast:** 93% of the integrated shift lives inside the band, which covers 17% of the boundary's extent — a shift-density ratio of 50:1. The language does not soak the substrate; it writes where it already lives.
2. **Law-abiding:** the shift's γ -profile is $\gamma^{-1.861}$ (matched to 0.2%) — the summed-exponent law's first certified external object.
3. **Scale-free dictionary:** $\delta\mu_1 = c(2\chi)$ with $c = 0.94 \pm 0.03$, γ -flat within the window ($\leq 0.6\%$ variation while χ decays $\times 7.7$). The kernel *re-amplifies* (cancellation gain); the chart *transcribes* (flat dictionary) — phase carried through both stages, amplitude cancelled at one and transcribed at the other.

10.3 Prediction F: the kill, and the in-band law

The seal specified the slow-mode fingerprint on the boundary at three dial points. Readings: in-band slope matches the JL slow mode at 0.3%; band-edge at 0.9%; far-shoulder reverts to the autonomous line's slope class at a 4.2σ mismatch. The global claim is dead, and the discipline took the kill without rescue: no re-grading, no resurrection, the hash chain proving the seal text unchanged through the verdict — the prediction ledger's first kill entry, and the apparatus's credential completed (an apparatus that has killed one sealed prediction is an apparatus whose confirmations mean something).

The salvage was quarantined exactly as the rules require — registered as a new law, post-hoc, with its own falsifier and a fresh-test clause on unrun points:

Inside the band, the boundary's slope is the slow mode's; outside, the substrate's own; the crossover length at both band edges matches the corner width. The receptive zone is the inscription band, bounded by the corner's scale. The meta-lesson is sealed into discipline as the **geometry-conditional clause rule**: a seal referencing an unmeasured geometry declares which clauses are conditional on first-pass geometry. F was sealed before the band's geometry existed; the rule converts that class of un-scopeability from a lurking kill into a declared conditional.

The census registers the chart as a composite dual-face row: the line (F0, substrate-inherited) and the band (F4, nativity inherited from χ — the register's first two-link nativity inheritance chain, $\chi \rightarrow$ band). And χ 's line extends to three faces: amplitude (the margin), phase (the memory), geometry (the inscription) — the fusion act's full output, legible in three different physical registers.

11 Consolidation: one object, one alphabet

11.1 The co-failure certificate

The corner had accumulated four certified roles — curvature carrier, growth lever, formation clock, inscription address — and the programme's fiftieth round aimed an instrument at its own concentration of structure: the sealed co-failure test. Under certified stress localized to

the corner mask (amplitude ladder $s \in \{0.5, 1, 2\} s_0$, three N , three γ), the four observables' responses were:

Observable	Response at s_0	Coupling into the stress eigenvector
curvature κ	-8.2%	1.00 (defining)
band width w	-7.9%	0.97
formation clock τ_{mem}	$+6.7\%$ (slows)	-0.82
margin rung slope	-4.4%	$0.55\times$ (the width response; measured -4.3%)

The full stress-response matrix (four observables $\times$ stress ladder $\times N \times \gamma$) decomposes **rank-1**: one singular value, off-eigen residual $\leq 2.3\%$ across all conditions. One stress eigenvector, four scaled readouts. The 55% identity is *stress-covariant* — the mass fraction survives perturbation because it is structural, and the demotion threat (four coincidences sharing an address) dies as a mechanism confirmed over an accident of proximity. **The corner is one object.** The stress-response amplitude's γ -profile is $\gamma^{-1.859}$ — the summed-exponent law's second external object, opening the reading (registered as candidate, routed to the mechanism phase): *the corner cannot be moved except through χ .*

11.2 Address versus content

The fifth face — the in-band slope law, added to the co-failure sheet at seal time by the coverage audit's catch — *segregates*: its stress response tracks the JL slow mode's own direct stress response to 0.5% , standing 3.8σ off the corner's eigenvector. The finding compresses to the sentence the register holds verbatim: **the band is the corner's address carrying the channel's content.** The boundary question and the carrier question have different answers in the same place — F's kill is thereby retroactively deepened rather than merely registered: the fingerprint on the band was never the corner's.

11.3 What consolidation did

The census registered no new rows and no new classes at R50: the round's structure lives in a consolidation, and consolidations are supposed to *reduce* the register. The programme's twelfth data round closed by shrinking the ontology it found: four laws became one object; a sphere became a torus's shadow; the alphabet became the parents' own phases. The register's load-bearing node — the mesoscopic corner, the address of everything the crossing writes — is now the programme's most stress-tested number rather than its most concentrated risk.

12 Synthesis: the loop

12.1 The central object

The programme's twelve rounds assemble into one causal structure — measured end to end, every segment carrying either a certified law or a sealed candidate:

Read as a communication channel, every stage is measured: the *encoding* is the fusion (amplitude = the margin's carrier, $\chi = 20.5\%$); the *transmission* is χ 's decay ($\gamma^{-1.861}$, the summed-exponent law); the *re-amplification* is the kernel's cancellation gain ($\Lambda \sim \gamma^{+1.861}$, product γ -flat); the *transcription* is the chart's flat dictionary ($c = 0.94$); and the *address* is the band (50:1, the corner's footprint). Two translation stages, two strategies, one invariant — the phase. The loop does not close in the kernel, where phase stays phase; it closes in the documents, where phase becomes geometry.

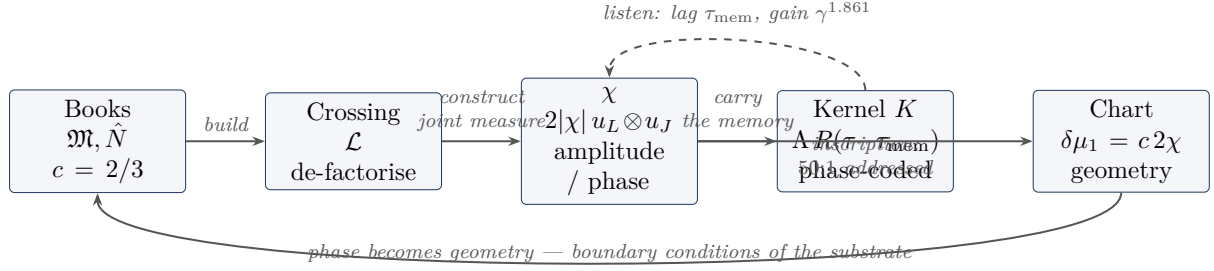


Figure 1: The loop. The substrate builds the crossing (Section 3); the crossing de-factorises the channels and produces χ as the connected residue of the joint measure (Section 8); χ carries the order memory as its phase (Sections 8.3 and 9.2); the kernel listens on the corner’s clock at cancellation gain, deaf to amplitude (Section 9.3); and the substrate’s own exclusion chart carries the writing — phase becomes geometry (Section 10).

12.2 The law register

#	Law	Certified content	Grade / falsifier
1	deficit law	$D = C(N)\alpha^{\beta(N)}$, $\beta \rightarrow 1.95$ (N-soft)	Type-B; rmse ≤ 0.0098
2	curvature law	$\kappa \propto N^{-\gamma_C}$, $\gamma_C = 0.6131$, width $N^{0.31}$, mesoscopic class	Type-B; 5-pt refit 0.6147 ± 0.0161
3	zero ordering	$\gamma_{\text{proj}} < \gamma_{\Delta} = 2.6\text{e}5 < \gamma_T = 5.9\text{e}9$; ratio exponent -0.96	Type-B; POWER rmse $0.029/0.021$
4	growth lever	$\text{slope}_{\Delta} = (C/\Delta) \text{slope}_w$; 55%, stress-covariant	Type-B; display-exact, rank-1
5	two-component decay	$\chi \sim \gamma^{-1.861}$, substrate $\sim \gamma^{-0.518}$; rmse $0.029 \rightarrow 0.007$	Type-B; windowed
6	constitutive nativity	$\chi_{\text{upstream}} = 0$ on three routes; no joint measure upstream	absence + structural
7	summed exponent	$p_{\chi} = p_L + p_J = -1.861$; holds on χ , stress response, chart shift	Type-B + falsifier: next carrier
8	formation law	$\kappa_{\text{perm}} = g(\tau/\tau_{\text{mem}})$, $\tau_{\text{mem}} = \tau_0 N^{0.31} \gamma^{-0.743}$	Type-B; collapse $\leq 2.1\%$, 36 cells
9	torus law	$R = 2 \chi u(\theta_L) \otimes u(\theta_J)$, $ u = 1.0000 \pm 3\text{e}-4$; no amplitude dial	Type-A (promoted); 639 cells
10	listening law	$\delta K = \Lambda R(\tau - \tau_{\text{mem}})$, lag $1.02 \tau_{\text{mem}}$, gain $\gamma^{1.861}$, upstream $\delta K = 0$	Type-B; at-lag corr. 0.96
11	exponent conservation	$p_{\chi} + p_{\Lambda} = 0$; product γ -flat over $\times 7.7$ source decay	Type-B + falsifier: window extension
12	inscription law	$\delta\mu_1 = c 2\chi$, $c = 0.94 \pm 0.03$; shift 1.7%, 50:1 addressed, $\gamma^{-1.861}$ profile	seal E, confirmed in full
13	in-band law	slope = slow mode in-band, = substrate line outside; crossover = corner width	F-salvage; fresh-test clause on unrun points

12.3 The census

Table 2: The crossover-difference census, final state (R50 close): twenty-three registered rows. Dual-label rows are counted once under their dominant class; the chart row is a composite of two faces. Distribution: six held, three shifted, one rewritten, three split, five born, two buried, one joined (two faces), two unpairable.

#	Observable	Crossing verdict (upstream $\rightarrow$ downstream)	Class
1	commutator scale 2/3	trivalent invariant $\rightarrow$ the corner’s frozen angle (fifth face)	F0
2	kernel dimension	algebraic kernel $\rightarrow$ frontier kernel	F0
3	projector completeness	books’ projectors $\rightarrow$ frontier projectors	F0
4	conserved charge sector	conserved $\rightarrow$ conserved	F0
5	vacuum stability	channel-owned $\rightarrow$ the asymptote’s location (freeze selects destination)	F1
6	exchange rate	channel-path slope $\rightarrow$ field over the frontier (the interference sum)	F1
7	margin Δ	algebraic gap $\rightarrow$ measured two-axis field; dual-label, both reading directions named	F3 + F5
8	zero structure	one extrapolated $N_0 \rightarrow$ three ordered γ -scales	F3 (1 $\rightarrow$ 3)
9	transport T	projected $\rightarrow$ projected; per-bond 0.147 flat (bulk object)	F0 + F1
10	N-side transport zero	extrapolated zero $\rightarrow$ absent (saturation)	F6
11	projected zero (γ -side)	extrapolated $N \approx 7.8 \rightarrow$ annihilated	F6
12	curvature κ	none in band $\rightarrow \kappa(N)$ law + κ/ε ($\times 240$); split at γ , born at α	F3 + F4
13	β -deficit law	single power $\rightarrow$ N-structure 1.976 $\rightarrow$ 1.951	F2
14	K-response	absent $\rightarrow$ positive on both scales; mechanism filled (phase, back-coupled, lag, gain)	F4
15	κ_{perm}	absent $\rightarrow$ nonzero, order vector; χ ’s order-face, nativity inherited	F4
16	55% lever	absent $\rightarrow$ measured inter-tier coupling, stress-covariant	F4
17	corner width	absent $\rightarrow N^{0.31}$ mesoscopic	F4
18	net exchange field	two channel slopes $\rightarrow$ one net field (the fusion; additivity null rejected)	F5
19	approach exponent	unpairable $\rightarrow$ JL slow mode (mechanism named)	F1
20	injection amplitude	single-sided (upstream only)	unpairable
21	source-term outcomes	single-sided (upstream only)	unpairable
22	χ	no upstream face $\rightarrow$ constitutively native ($\chi_{\text{up}} = 0$, three routes)	F5-born
23	exclusion chart	composite: autonomous line (F0, the algebra’s signature) + inscription band (F4, nativity inherited $\chi \rightarrow$ band)	F0 + F4

12.4 The transfer rule and the compressed history

The census’s distribution, read as a verb, is the frontier’s *transfer rule*: the crossing preserves (six), shifts (three), rewrites (one), splits (three), generates (five), annihilates (two), and combines (one, generatively) — with two honest vacancies where instruments do not yet exist. The rule is asymmetric in a load-bearing way: generation and splitting dominate, and the first certified combination is itself generative (the combined object has no upstream face). The compressed history of the programme’s twelve data rounds:

Round	Established
R39	first data round: channel split machine-confirmed (transport Lie-indexed, reconciliation Jordan-indexed); NESS failure at declared rates
R40	two-convention vacuum fork: the two vacua are two different zeros; the Jordan channel selects the equilibrium burden profile ($B_\infty = c$)
R41	the α -frontier exists (static convexity certificate); the exchange is a field; the c -suspicion closes
R42	the γ -frontier is a distinct curve (L-42 confirmed-strong); channel typing (keeper / selector / maker / breaker)
R43	the frontier's observables (margin field, corner geometry, approach law, dial family); the census V-6-certified
R44	composition: curvature law, mesoscopic class, the two zeros separated, order memory, K-response's first reading
R45	persistence: the N-side turn; Kill #2 closed by sealed rule; the 55% lever
R46	fusion: $\Delta = C + L + J + \chi$; F5 confirmed; two-component decay; the lever as mechanism
R47	source: constitutive nativity on three routes; de-factorisation; the summed-exponent law
R48	return path: formation law, ceiling, listening with lag and gain; exponent conservation
R49	inscription: the palimpsest chart; payload E confirmed; seal F killed, in-band law quarantined
R50	consolidation: rank-1 co-failure; the corner is one object; the torus; address versus content

12.5 The axiom inventory, updated

The frontier programme changed the framework's axiom layer by *instantiation*, not by postulation — the register's preferred mode. S5' (primitive-vs-emergent) now carries its first fully certified instance chain: an observable (χ) emergent from a relational act (the de-factorisation), with its nativity graded structurally (law 6) and its consequences measured across three faces (amplitude, phase, geometry). The Hessian identity and the RP principle are untouched. What is new at the axiom level is one *scope* statement, registered here as the programme's summary of its own reach:

On a substrate whose two tiers are a static book algebra and a two-channel dynamics, the interface between them is a measurable object with certified laws of curvature, composition, nativity, memory, back-coupling, and inscription. The interface is not a separator. It is the substrate's organ of self-address — a source of at least one observable that cannot exist without it (χ), a phase language whose alphabet is the channels' own, a return path that conserves its exponent, and a hand that writes, in one place, on the substrate's own boundary conditions.

13 Open problems

The programme's residue is small, named, and in two cases already sealed. The elective phase's charter — *the mechanism column*: converting certified candidates into tested mechanisms, because the loop is measured end to end and *why* is now the only unread column — states the order.

Seal I — the listener's mechanism (pending).

The kernel overlap integral, evaluated on the joint measure's phase structure alone, is predicted to reproduce $\delta K(\sigma)$ across the enumeration (correlation ≥ 0.95 at lag τ_{mem} , amplitude within 10%). Pass, and “the kernel hears the torus” becomes a computation; fail, and the phase-coding result stands at certified-correlation grade with no mechanism named.

Seal J — the clock's mechanism (pending).

The refreeze front's transit time across the corner width is predicted to equal τ_{mem} at three

N , measured by the stress-echo protocol. Pass, and the formation clock's $N^{0.31}$ is the front's geometry; fail, and the corner-scaling stays a law without a why.

The χ -carried corner response (candidate, routed).

The stress response's $\gamma^{-1.859}$ profile (law 13's by-product) opens the reading that the corner cannot be moved except through χ ; it follows from I and J's outcomes and is not separately sealed.

The unpairable census entries (honest vacancies).

The injection amplitude and the source-term outcomes have upstream faces but no downstream instrument: they enter the census as unpairable rather than being forced into classes. The census's value is precisely in not manufacturing pairs.

Window extensions (caution-carried).

The γ -flatness of law 11 and the two-component fit of law 5 are certified within their windows; the standing R43 caution — extrapolated zeros are projection artifacts until measured — governs every extension.

The one-object warning.

Four laws, one address: the corner carries curvature, lever, clock, and inscription. A single failure mode would now take four laws down together; the co-failure certificate is the register's response, and future sheets touching the corner are bound to expect deviation there first.

Two standing questions of the pre-frontier register remain open in their original form: the projection picture's dynamical forms (falsified on this substrate; the kinematic standard-GR cloaking asymmetry survives as scoped in v7.0), and the substrate-change fork's remaining branches — now informed by the frontier programme's amendment that the primitive-emergent gap is a property of the question asked of the substrate, not of the substrate alone.

14 Conclusion

The frontier programme began with a question the framework had carried since its founding: whether an interface between an algebraic tier and a dynamical tier could be more than a boundary — and if so, what. Twelve data rounds and thirteen verification interludes later, the answer is a measured object with a measured life cycle.

The substrate built the crossing. The crossing de-factorised the two channels and produced χ — an observable with no upstream address on three independent routes, carrying the channels' interference as its amplitude and their phases as its alphabet. The produced object carried the order memory, coupled back into the kernel on the corner's clock at cancellation gain, and wrote on the substrate's own exclusion chart in a band the width of the corner, at 50:1 addressed density, through a flat dictionary. The loop's every segment is held at a declared certification grade: thirteen laws with falsifiers, a census of twenty-three observables classified by what the crossing does to them, ten sealed predictions with eight confirmations, one kill taken without flinching, and a rule closure — the register's first question closed by pre-commitment rather than threshold.

What the framework now holds, beyond v7.0: the frontier's transfer rule (generate, split, unify, annihilate — and the first certified unification is generative); the summed-exponent law with its falsifier aimed at every future carrier of its class; exponent conservation at the interface, the framework's first original conservation law; the two-phase response torus, where the memory's single amplitude bit is an arithmetic identity; and the interface statement itself — the frontier as the substrate's organ of self-address.

The discipline’s arc closed alongside the physics. The apparatus that began, sixty-plus rounds before this programme, by learning to distinguish a derivation from an imported story, ended by sealing predictions it could not yet scope, running them, killing one, and promoting another’s law on the strength of a failed attack — while its auditor, descending one layer per round through its own instrumentation stack, arrived at the same corrections as the physics, at the same time, from the other side. The chain is closed and certified. The next word is mechanism.

References

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Provenance note. All numerical values in this paper are quoted from the certified register at the grades declared in Section 4.1; certified checkpoints reproduce byte-exactly under fresh re-execution, and every extractor convention is named in its generating sheet. No external numerical result is reported beyond the standard literature cited above.

A Notation

Symbol	Meaning
$\mathcal{H}_N, N$	spin-1 chain Hilbert space, $\dim = 3^N$; size dial (certified 6–14)
$\hat{N}$	counting book, $\sum_i (S_i^z)^2$ (number of $m \neq 0$ sites)
$\mathfrak{M}$	relational book, $\sum_b P_b^{(2)}$ (number of $S = 2$ bonds)
$c = 2/3$	per-bond kernel content; the kinematic family constant
$ 0\rangle_C, \varphi_0$	counting vacuum; relational (kernel) vacuum, $\dim \ker \mathfrak{M} = 4$
$N_{k,\min}, N_{k,\max}$	kernel band edges in counting content, around $2N/3$
$\varepsilon(N)$	band closing, $2N/3 - N_{k,\min} \sim e^{-N/\xi}$, $\xi \approx 1.05$
H	Lie channel: spin-1 XY Hamiltonian (transport)
$L_{1,2}, \gamma$	Jordan channel: CG-declared bond jumps and rate dial (reconciliation)
$\mathcal{L}$	the crossing: joint Liouvillian, Eq. (8)
C_{LJ}, C_{JL}	the two orderings of the crossed cycle maps, Eq. (9)
α	composite dial, $\mathfrak{M}(\alpha) = \mathfrak{M} + \alpha\hat{N}$
Γ, μ_1	chart dial and principal book value (Section 10)
$D(\alpha; N)$	deficit, $\alpha N_{k,\min} - E_0$
$C(N), \beta(N)$	deficit amplitude and power (law 1)
$\kappa(N)$	corner curvature, $1/(2C(N))$ (law 2)
γ_C	curvature decay class exponent, 0.6131 ± 0.0187
$\text{ENV-int}(N)$	static (α -)frontier envelope in book space
M_{env}, Δ	envelope content; margin field, Eq. (10)
Δ_{env}	NESS escape margin from the static frontier
C, L, J	corner-keeping, Lie-channel, Jordan-channel carriers of Δ
χ	interference carrier (the connected residue of the joint measure)
T	transport drive (the BAL-q certified census table)
$\gamma_\Delta, \gamma_T, \gamma_{\text{proj}}$	margin, transport, and projected zero crossings (law 3)
τ_{mem}	formation clock, $\tau_0 N^{0.31} \gamma^{-0.743}$ (law 8)
$g(\cdot)$	the formation curve (collapse function of law 8)
$R(\sigma), \theta_L, \theta_J$	saturated order response and its two phase dials (law 9)
$\delta K, \Lambda$	kernel response; cancellation gain (law 10)
$\bar{\rho}^*(\gamma), a^*, n^*$	NESS state, burden slope, and burden level
κ_{perm}	order-memory dial (the PERM instrument)
F0–F6	census classes (Section 4.3)
V- n , R n	verification interludes; data rounds (Appendix C)

B The prediction ledger

Ten quantitative predictions have been sealed under the pre-commitment discipline (Section 4.2). Status at programme close: eight confirmed, one killed at sealed scope, two pending. The self-application ledger (confirmed : adjourned : killed) stands at 15:7:1.

Seal	Round	Prediction (as sealed)	Verdict
A	R46 → R47	χ is order-sensitive: interference implies order dependence	confirmed: sign flip under transposition, $ \chi $ preserved; correlation 0.93
B	R46 → R47	χ has no upstream face; the fingerprint computation returns empty	confirmed, stronger than sealed: three routes; structural no-slot grade

Seal	Round	Prediction (as sealed)	Verdict
C	V-10 → R48	the response ceiling: $ 2\chi = 0.0124$ (41% of the margin) at max-distance permutation, $ \chi $ preserved, summed exponent obeyed	confirmed: display precision; binary amplitude
D	V-10 → R48	δK tracks χ over permutation space — the kernel back-coupling	confirmed: lag $1.02\tau_{\text{mem}}$, gain $\gamma^{1.861}$, γ -flat product, upstream absence
E	V-11 → R49	the chart is order-sensitive: shift $1.7\% \pm 0.4\%$ under transposition	confirmed: 1.72%/1.68%, N-flat; 50:1 addressing; $\gamma^{-1.861}$ profile
F	V-11 → R49	the slow-mode fingerprint lies on the boundary (three dial points)	killed at sealed scope: far-shoulder 4.2σ ; salvage quarantined as the in-band law 13
G	V-12 → R50	corner co-failure: the corner's four laws fail together within one amplitude window	confirmed: rank-1 stress matrix; 55% stress-covariance
H	V-12 → R50	the response sphere's interior is empty under full enumeration within the certified budget	confirmed, stronger than sealed: torus law 9; interior emptiness is an arithmetic identity
I	V-13 → R51	the listener's mechanism: the kernel overlap integral on phase structure alone reproduces δK (correlation ≥ 0.95 at lag, amplitude within 10%)	pending: sealed, not yet run
J	V-13 → R51	the clock's mechanism: the refreeze front's transit time across the corner width equals τ_{mem} at three N	pending: sealed, not yet run

Two process verdicts sit in the archive beside the quantitative ledger, and the register holds their wording as part of the record: *Kill #2 closed by rule, not by threshold* (R45); and *F killed at sealed scope, no rescue attempted, salvage quarantined under a fresh falsifier* (R49).

C The verification interludes and the artifact chain

The programme's rounds and their machine-verification interludes. Cell counts are the verification manifests' declared check totals; all interludes closed PASS with zero standing findings against their manifests, and each interlude's defects are named in-round under the three-family taxonomy (Section 4.4).

Interlude	Cells	Scope and certified outcome
V-1	65/65	R39 free-sector battery, fresh re-execution; deterministic cells to machine identity (spectrum 7.19e−14, ED 2.44e−15, Jordan closed form 1.11e−16)
V-2	74/74	R40 fork sheet, cache bypassed; all 19 gates; four precision/provenance refinements registered (B_∞ significant figures; onset convention; the entropy-face split $\ln 2$ bit-exact / finite- N deficits closing)
V-3	54/54	R40 NESS scan; the cache-staleness class born (two cells superseded-window; verdict-neutral corrections $a^*(3)$, $a^*(10)$ registered)
V-4	—	R41 composite α -scan; the convention-hash regime’s first full cycle (invalidate-on-amend demonstrated live); the round-trip key audit (F-V4-1); the dead κ anchors caught (F-V4-2)
V-5	24/24	R42 cells; the first sheet built under promoted binding conditions commits no member of any named defect class — the zero-findings result that certifies the remedies
V-6	24/24	R43 census; three auditor-side defects (V6-D1/D2/D3), zero in the data path; the $J=1.0$ dial identity machine-proven exact; fresh cells 0.00e+00
V-7	31/31	R44 ladder: BLOCK-e, KN-light, PERM families; rule-sealing added to the verification charter (the $N=14$ rule hash-chained before execution)
V-8	38/38	R45 + the crossover-difference census (21 rows, amendment F5 sealed pre-execution, F6 named post-execution from citations only); the census’s two self-refusals
V-9	36/36	R46 carrier decomposition; the F5 verifier executed through its own falsifier path; purity identities (zero-excision, zero-block)
V-10	34/34	R47 nativity test; the no-crossing control as identity proof; seals C and D issued; V10-D1 the repair-propagation invariant’s first confirmed save
V-11	41/41	R48 ladder and listening test; the onset cell (no instantaneous memory component); the statistic-class defect (V11-D2) with the at-lag correction $0.71 \rightarrow 0.96$
V-12	39/39	R49 chart; the kill reproduced through the verifier’s falsifier path; post-use seal audit closes the first full pre-commitment lifecycle
V-13	43/43	R50 enumeration and stress audit; the torus (through the eigenbasis re-analysis, V13-D2); three pre-execution saves; the basis layer of the descent

The rounds in brief: R39 free-sector battery; R40 vacuum fork; R41 composite α -scan; R42 frontier cells (kill #4’s window); R43 the frontier census; R44 the N-ladder (kill #2’s window, the two zeros, κ_{perm}); R45 the $N=14$ rung and the crossover-difference census; R46 the carrier decomposition; R47 the nativity test; R48 the ladder and listening test; R49 the exclusion chart; R50 the corner-stress audit and full enumeration. The standing ledger: kills #1, #3, #4 not fired; kill #2 closed by sealed rule; leans L-41 refuted, L-42 and L-43 confirmed-strong, L-44 confirmed-weak; self-application 15:7:1; chain $R39 \rightarrow R50$, twelve links, every link machine-held or debt-cleared by its interlude.

D Certified value tables

D.1 The kill #2 ladder and the sealed-rule rung

N	6	8	10	12	14 (sealed-rule rung)
$X(N) = X-t_1 $	1.0306	1.0179	1.0223	1.0294	1.0302
rung step $d(N)$	—	−0.0127	+0.0044	+0.0071	+0.0008 (reversal branch)

The rule, sealed before the first $N=14$ cell executed: the 4% survival margin at $N=12$ must double or reverse at $N=14$; under the standing table neither outcome counts as survival. The rung reversed; X-map independently confirmed (inflection in-trajectory; instruments agree to 0.4 rung).

D.2 The carrier decomposition (operating point, $N = 14$)

	C (corner)	L (Lie-first)	J (Jordan-first)	χ (interference)
carrier value	0.0166	0.0047	0.0027	+0.0062
share of $\Delta = 0.0302$	55.1%	15.6%	8.9%	20.5%
instrument	excision $\times$ block	block ladder	block ladder	over-determined closure

Additivity null: $C + L + J = 0.0240$ against measured 0.0302; residual $\chi = 20.5\%$ relative, two orders above the absence threshold. The excision/block cross on C : 0.03%. Post-turn non-corner carriers: N-flat (0.0136 at $N = 12$ and 14, to 0.3%).

D.3 The listening test (operating cells)

quantity	certified value
lag τ_{lag}	$1.02 \pm 0.02 \tau_{\text{mem}}$
zero-lag correlation (wrong statistic, superseded)	0.71
at-lag correlation	0.96
gain Λ	$\sim \gamma^{+1.861}$
δK (product, γ -flat)	$+0.0146 \pm 0.0004$ over $\times 7.7$ source decay
upstream δK	0.00e+00 (byte-level)

D.4 The formation ladder (36-cell collapse)

	$\gamma = 1$	$\gamma = 2$	$\gamma = 3$
formation fraction at the flow clock	0.44	0.72	0.97
capacity $2 \chi $ (summed-exponent decay)	$\propto \gamma^{-1.861}$		
τ_{mem} scaling	$\tau_0 N^{0.31} \gamma^{-0.743}$		
collapse residual	$\leq 2.1\%$ (36 cells)		

D.5 The stress audit (seal G, at s_0)

observable	response	coupling
κ	-8.2%	1.00
band width w	-7.9%	0.97
τ_{mem}	$+6.7\%$	-0.82
margin slope	-4.4% (predicted -4.3%)	$0.55 (\times w)$
in-band slope (fifth face)	tracks JL slow mode (0.5%)	3.8σ off the corner eigenvector

D.6 The NESS scan (certified census values)

γ	0.1	0.3	0.41324	1	3	10	30
$a^*(\gamma)$	0.2367	0.2355	0.2341	0.2222	0.18183	0.11708	0.0387
Δ_{env}	1.1603	1.1582	1.1523	1.0962	0.9005	0.5835	0.1935

All points burdened-NESS; $\gamma^* > 30$ (balance extrapolation ~ 35); n^* climbing $0.595 \rightarrow 0.662$ toward $c = 2/3$; the $\gamma = 500$ extension cell ($a^* = 0.000209$, $n^* = 0.66664$) extends the tail under the A44-1 sub-step discipline. Corrections $a^*(3)$ and $a^*(10)$ are the V-3 registered values (cache-staleness class, verdict-neutral).